Effective Communication in Recruitment Processes for Entry-Level Computing Professionals: A Systematic Mapping Study

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Abstract — **In the contemporary knowledge economy, effective communication represents a strategic competency for Information Technology (IT) professionals, yet a persistent "soft skills gap" often hinders entry-level practitioners during recruitment processes. This Systematic Mapping Study, conducted according to the Kitchenham and Charters protocol, investigates how artificial intelligence (AI)-driven tools and structured methodologies can enhance the communicative performance of novices. By analyzing fifteen high-quality studies retrieved from the ACM Digital Library, IEEE Xplore, and Scopus databases published between 2020 and 2026, the research addresses the urgent necessity of bridging the disparity between academic theoretical training and professional industry requirements. The findings reveal a sophisticated technological ecosystem, incorporating large language models, natural language processing, and multimodal affective computing, that facilitates immersive mock interviews, real-time feedback, and objective behavioral assessments. These interventions effectively mitigate evaluative anxiety and rectify self-assessment deficits, although critical challenges regarding algorithmic transparency and human empathy persist. The review concludes that while technology significantly optimizes workforce readiness and reduces human bias in screening, sustainable integration requires a multidimensional approach addressing psychological, cognitive, and structural barriers. Consequently, future research should explore underrepresented geographic regions and refine the synergy between automated feedback and human mentorship to ensure graduates are strategically positioned for the global talent market.**

Keywords—Effective communication, recruitment processes, entry-level IT professionals, AI tools, communication strategies.

I. Introduction

In the contemporary knowledge economy, effective communication serves as the fundamental driving force behind professional success and seamless organizational integration for Information Technology (IT) professionals [1][9]. Beyond mere interpersonal interaction, communication functions as a critical interface that transforms abstract technical potential into actual business value by aligning technological innovation with strategic corporate objectives [11]. The ability to articulate complex logic, negotiate requirements, and collaborate within multidisciplinary teams is what ultimately determines the impact of a professional's technical expertise [1][9].

However, the computing industry currently faces a persistent paradox. While technical excellence (hard skills) is rigorously cultivated throughout academic training, it is frequently overshadowed by a profound lack of interpersonal and communicative proficiency—a phenomenon often referred to as the soft skills gap [1][6]. This disparity is particularly evident among entry-level IT professionals and new graduates, who often possess high levels of theoretical knowledge yet find themselves practically underprepared for the performative and communicative demands of modern selection processes [12][17]. Foundational research has long highlighted a critical curricular imbalance, noting that academic perceptions often prioritize technical expertise over interpersonal development [5]. Conversely, contemporary industry studies emphasize that collaborative skills have transitioned from desirable traits to non-negotiable requirements for employment [4].

Functioning as a transformational engine in this landscape are structured communication strategies and AI-driven communication tools. Advances in Large Language Models (LLMs), natural language processing (NLP), and multimodal affective computing are redefining how junior professionals navigate the transition from student life to professional work [10][11][20]. These technologies provide immersive, real-time simulated environments that offer personalized feedback on verbal fluency, emotional tone, and non-verbal cues, enabling a more data-driven approach to career construction [3][17]. By providing a safe, repeatable space for practice, these tools mitigate the high levels of anxiety and lack of confidence that frequently hinder novices during technical interviews,,.

The motivation for this research stems from the significant academic gap regarding soft skills training within higher education institutions. While the industry demands holistic professionals, many university programs continue to prioritize theory-heavy curricula, leaving students to discover the importance of communication skills only when they face the challenges of the job market [6]. Investigating the intersection of communication techniques and AI-mediated training is therefore vital not only for the personal development of graduates but for their strategic positioning in an increasingly competitive global talent pool (Nirgude and Awasekar, 2025).

This study is structured as a Systematic Literature Review (SLR), following the established methodological guidelines of Kitchenham and Charters [7]. The review aims to investigate how the strategic integration of AI-driven tools and structured communication methodologies is transforming the recruitment landscape for novices. Specifically, this SLR addresses how efficient communication can be enhanced through technology to bridge the industry-academia gap and ensure that inexperienced professionals can successfully demonstrate their potential during selection processes.

This paper is divided into six sections. Section 2 presents the background and related work concerning the soft skills gap and the evolution of AI in recruitment. Section 3 details the SMS methodology, including the research questions and data extraction protocols. Section 4 provides the results and a comprehensive analysis of the identified studies. Section 5 discusses the practical implications for Computer Science education and the IT industry, while Section 6 concludes with final findings and recommendations for future research.

II. Background

In the contemporary Information Technology (IT) landscape, effective communication is defined not merely as a peripheral soft skill but as a strategic competency essential for professional integration and the transformation of technical knowledge into organizational value [16]. Organizations increasingly recognize that the ability to articulate complex logic and collaborate within multidisciplinary teams is a primary determinant of a professional's success [15]. However, a persistent soft skills gap remains a defining challenge in Computer Science education, where graduates frequently possess high levels of technical proficiency (hard skills) yet remain practically underprepared for the communicative demands of the professional workforce [9]. This disparity is central to the findings of foundational research such as Everything but Programming, which highlights the lack of interpersonal development in undergraduate curricula, and Habilidades Colaborativas no Mercado de TI, which emphasizes that the industry now views these competencies as essential requirements for employment [9].

Artificial Intelligence (AI) has emerged as a set of sophisticated technologies designed to simulate human capabilities, including learning, pattern recognition, and adaptive reasoning, providing a powerful mechanism to bridge this industry-academia gap [12]. Within the context of selection processes, AI is applied through Natural Language Processing (NLP) to conduct automated interview analysis, sentiment detection, and coherence assessments of candidate responses [17]. Advanced mock interview simulators, such as VoxMentor and the IIAA platform, leverage large language models (LLMs) to create immersive, real-time simulated environments that mirror real-world dynamics [10][11]. These systems provide AI-powered feedback on both verbal fluency and non-verbal cues, mirroring the functions of real-world analogs such as the recruitment-driven AI platforms used by major enterprises like Boss Zhipin to optimize person-job matching [6][20].

The intersection of technology and behavioral performance allows for the enhancement of structured communication methodologies, such as the STAR (Situation, Task, Action, Result) method, through data-driven feedback loops [6]. By utilizing techniques like Think-Aloud practice, students can develop the ability to verbalize complex thought processes while simultaneously executing technical tasks, a dual demand that often induces high levels of anxiety [2]. Technological interventions offer significant potential for enhancing efficiency and ensuring a more objective evaluation by reducing human bias in the initial screening stages [17]. Nevertheless, critical challenges persist, including concerns regarding algorithmic transparency, the potential for algorithmic bias, and a lack of human empathy in providing feedback on deep-seated anxieties [13]. Addressing these barriers is vital for the responsible integration of AI into career development, ensuring that technology serves as a "buddy" rather than a "crutch" for inexperienced professionals navigating the complexities of the modern job market [6].

III. Research method

This study adopts the Systematic Mapping Study (SMS) approach, following the guidelines proposed by Kitchenham and Charters [7], structured in three main stages: planning, conducting, and reporting. The SMS methodology is particularly suited for broad research domains where the goal is to identify, classify, and synthesize the body of available evidence, providing an overview of the research landscape rather than focusing on a specific narrow question. This approach was chosen given the multidisciplinary nature of the research topic, which intersects communication skills, human-computer interaction, artificial intelligence, and technology-enhanced learning.

A. Research Planning

The planning stage involved defining the research questions, selecting appropriate data sources, and establishing the search and selection protocol. The central research question guiding this study was formulated as follows: *"How can effective communication of entry-level IT professionals be enhanced in recruitment processes through the strategic use of methodologies, techniques, and technological tools?"*

This central question was further detailed through four secondary questions:

- Q1: What AI-powered technological tools can be used in IT recruitment processes?
- Q2: What are the main communication barriers and failures faced by entry-level professionals in IT recruitment processes?
- Q3: What positive and negative impacts do AI-powered technological tools and communication strategies produce in IT recruitment processes?
- Q4: What structured communication techniques and strategies are most effective for entry-level technology professionals?

The selection of databases prioritized those most representative of the research domain in computing and educational technology. Three widely recognized academic databases were chosen: ACM Digital Library, IEEE Xplore, and Scopus. These repositories are well-established sources for peer-reviewed publications in the fields of computer science, information technology, and human-computer interaction, ensuring broad coverage of studies directly aligned with the investigated topic.

The search strategy was developed through a carefully constructed query string, incorporating key terms related to the research topic and their corresponding synonyms. The query definition process involved successive refinements conducted collaboratively by the research team until the initial results adequately reflected the scope of the investigation. Key concept groups included: (i) communication skills and interview preparation; (ii) recruitment and technical interview processes; (iii) AI-powered tools and simulation technologies; and (iv) entry-level computing professionals. The final search string applied across all databases was:

> *("communication skills" OR "interview preparation" OR "interview coaching") AND ("technical interview" OR "job interview" OR "hiring process" OR "recruitment") AND ("AI tools" OR "automated feedback" OR "interview simulation" OR "mock interview" OR "AI-powered" OR "natural language processing") AND ("junior" OR "early-career" OR "entry-level" OR "IT graduates" OR "computer science students") AND (PUBYEAR > 2020 AND PUBYEAR < 2027)*

To capture the most recent developments in AI-assisted communication training and recruitment technologies, the search was restricted to publications from 2020 to 2026. Studies published outside this range were excluded from the initial results.

B. Research Conducting

Executing the search protocol yielded a total of 233 studies in the initial retrieval: 131 from ACM Digital Library, 44 from IEEE Xplore, and 58 from Scopus. All results were exported and organized in a spreadsheet for systematic filtering and analysis. The study selection process was conducted rigorously to ensure that only the most relevant articles were included in the final analysis.

The selection followed a predefined set of inclusion and exclusion criteria, applied in sequential filtering stages.

The selection process was conducted in two stages. In the first stage (Filter 1), each article was assessed based on its title and abstract. Studies that clearly aligned with the research scope were retained, while those with evident thematic misalignment were excluded. This stage resulted in 8 articles from ACM, 21 from IEEE Xplore, and 23 from Scopus, totaling 52 studies for further analysis. A second filter examining the introduction and conclusion was not deemed necessary, as the first filter proved sufficiently discriminating for most articles.

C. *Quality Assessment*

To ensure that the selected studies were methodologically sound and made meaningful contributions to the research questions, a quality assessment was conducted based on five predefined criteria, aligned with recommendations by Kitchenham and Charters [7]:

- QC1. Methodological clarity: the study presents a well-described research method;
- QC2. Clear application context: the study defines the context of its application;
- QC3. Well-defined model or proposal: the study presents a clear contribution artifact or framework;
- QC4. Discussion of results: the study provides a substantive analysis and interpretation of findings;
- QC5. Acknowledgment of challenges, limitations, or threats to validity.

Each criterion was scored on a standardized scale: 1.0 for full compliance, 0.5 for partial compliance, and 0.0 for non-compliance, based on a surface-level review of each article's structure, abstract, introduction, and conclusion. A minimum quality threshold of 60% of the total possible score (i.e., at least 3.0 out of 5.0 points) was established for inclusion in the final analysis. Only studies meeting this threshold were retained for evidence collection and synthesis.

After applying the quality criteria, 15 studies were selected for the final analysis: 4 from ACM Digital Library, 5 from IEEE Xplore, and 6 from Scopus. Table 1 summarizes the complete selection process across all stages.

Table 1: Study Selection Process Summary

Source	Initial Retrieval	After Filter 1 (Title & Abstract)	After Quality Assessment	Final
ACM Digital Library	131	8	4	4
IEEE Xplore	44	21	5	5
Scopus	58	23	6	6
Total	233	52	15	15

D. *Data Extraction and Synthesis*

Following the quality assessment, data extraction was conducted systematically to consolidate relevant information from each selected article. Systematic mapping studies require a structured classification scheme to organize publications in a meaningful way, ensuring that contributions are aligned with the research questions [7].

For each retained study, extracted evidence included: AI tools and technologies referenced, communication barriers identified, reported benefits and drawbacks of tools and strategies, and specific techniques for communication improvement in interview contexts. All extracted data were recorded in a spreadsheet, where each row corresponds to a study and columns contain bibliographic metadata (title, year, authors, database, page count), quality assessment scores, and responses to each of the four secondary research questions (Q1 to Q4).

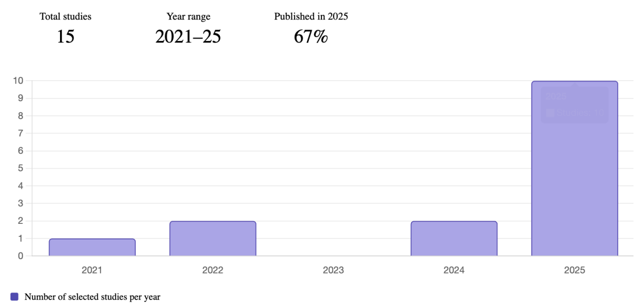
Each study was evaluated on how well it addressed each secondary question, using a three-point scale (1 = fully addresses, 0.5 = partially addresses, 0 = does not address). The extracted data were then categorized through thematic analysis, with recurring patterns identified and consolidated across studies. Studies could contribute to multiple categories, and items not addressed were labeled as "No response." This process enabled the identification of thematic clusters supporting a structured synthesis of the main findings related to communication enhancement in IT recruitment processes for entry-level professionals.

IV. Report and Discussion

A. *Studies Overview*

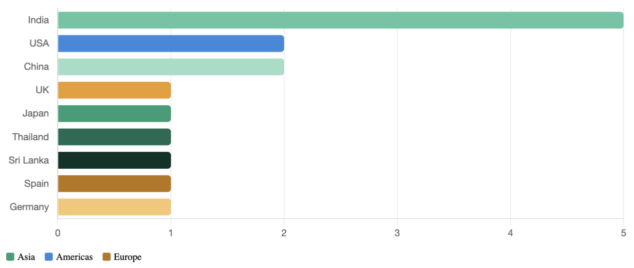
The systematic mapping process resulted in 15 selected studies (PS01–PS15), spanning the period from 2021 to 2025. The temporal distribution of publications reveals a pronounced concentration in 2025, with 10 studies published in that year alone. This trend reflects the growing intersection of artificial intelligence and educational technology applied to recruitment preparation, particularly following the widespread adoption of large language models (LLMs) and generative AI tools from 2023 onward. The remaining studies are distributed across 2021 (1 study), 2022 (2 studies), and 2024 (2 studies), suggesting that the field has gained significant momentum only recently.

Bar chart showing annual distribution of selected studies from 2021 to 2025



Regarding geographical distribution, the corpus shows a dominant presence of studies conducted in Asian countries, with India leading the representation (33.3%), followed by China (13.3%). Other countries represented include the United States (13.3%), Japan (6.7%), Thailand (6.7%), Sri Lanka (6.7%), Spain (6.7%), Germany (6.7%), and the United Kingdom (6.7%). Notably, no studies were identified from Latin America or Africa, which points to a significant geographic gap in research addressing communication skills and interview preparation for entry-level computing professionals. This concentration in South and East Asia may be attributed to the rapid growth of IT labor markets in these regions, government-driven investment in AI-powered education, and the increasing demand for workforce readiness tools among large populations of computing graduates.

Horizontal bar chart showing geographic distribution of studies by country



Concerning publication venues and study types, the corpus is composed predominantly of conference papers (80%), reflecting the dynamic and rapidly evolving nature of this research area, where new developments are disseminated quickly through academic events. Journal articles and full articles account for the remaining portion of the corpus (20%). The primary sources include IEEE Xplore (5 studies), Scopus (6 studies), and ACM Digital Library (4 studies), representing a balanced coverage of the main computing and information systems research repositories.

In terms of methodological approaches, the selected studies employ a diverse set of research methods. Several studies adopt a system development and evaluation approach, proposing AI-powered tools or platforms and assessing their effectiveness through user studies or automated metrics (PS01, PS02, PS06, PS08, PS09, PS14). Others rely on experimental designs involving human participants in controlled settings, such as virtual reality environments or real-time feedback systems (PS04, PS10, PS13). Survey-based and framework-oriented methods are also present (PS03, PS07, PS15), alongside text mining and data analysis techniques applied to job postings or educational corpora (PS12). One study follows a design-based research approach centered on conversational AI for technical interview practice (PS05). This methodological diversity reflects the interdisciplinary character of the field, bridging human-computer interaction, natural language processing, educational technology, and organizational behavior.

B. *Existing AI Tools and Technologies for Interview Preparation*

In the context of “Artificial Intelligence Tools Applied to IT Recruitment Processes,” this section addresses the research question “Quais ferramentas tecnológicas com IA podem ser usadas em processos seletivos de TI?” (What AI technological tools can be used in IT selection processes?).

The literature demonstrates that modern IT recruitment relies on a sophisticated ecosystem of artificial intelligence systems. These technologies prominently include Large Language Models (LLMs) and Natural Language Processing (NLP) for dynamic dialogue generation and resume parsing, Convolutional and Recurrent Neural Networks (CNNs/RNNs) for the multimodal analysis of behavioral and emotional cues, text mining algorithms for labor market requirement extraction, and immersive Virtual Reality (VR) integrated with microlearning frameworks. Collectively, these tools are deployed across the entire hiring funnel—from pre-selection screening and automated person-job matching to interactive mock interviews and objective performance scoring—fulfilling the critical roles of reducing human bias, scaling candidate assessment, and bridging the persistent gap between academic skill sets and rigorous IT industry demands.

In the initial stages of the selection process, AI tools are heavily utilized for pre-selection analytics, resume screening, and person-job matching. The study identified as PS12 presents a text mining and topic modeling approach, specifically Latent Dirichlet Allocation (LDA), applied to LinkedIn job advertisements to uncover industry-demanded hard and soft skills. Within the IT selection process, this method is used to structurally group professional requirements, fulfilling the role of market analysis and curriculum alignment to ensure candidates possess the exact competencies recruiters seek. Building upon this matching phase, PS06 introduces an advanced mock interview platform that utilizes NLP and pre-trained LLMs (such as BERT and T5) combined with Retrieval-Augmented Generation (RAG). This tool is used to parse resumes and compute similarity scores against job descriptions, fulfilling the role of automated candidate screening and targeted skill matching prior to the interview stage. Furthermore, PS08 details the MockLLM framework, which employs a multi-agent behavior collaboration paradigm where LLMs simulate both the interviewer and the candidate. Used during the post-interview review stage, this tool relies on reflection memory and a "handshake protocol" to evaluate mutual preferences, fulfilling the role of dynamic two-sided matching and final hiring decision-making.

Transitioning to the assessment phase, conversational AI and generative LLMs are widely applied to simulate realistic IT interviews and dynamically evaluate candidate responses. PS01 presents an AI-powered virtual simulator leveraging GPT-4 and real-time facial recognition. This tool is used to intelligently parse resumes and adaptively generate personalized questions, fulfilling the role of automated preliminary screening, interactive interviewing, and biometric anti-fraud security. Similarly, PS02 introduces an Interactive Interview Assistant Application that combines MetaHuman technology, ChatGPT 3.5, and Speech-to-Text systems to animate a digital avatar. It is used to immerse candidates in specific IT scenarios (e.g., Software Engineering), fulfilling the role of dynamic dialogue generation and formative candidate assessment. To further augment this stage, PS03 offers a framework integrating the Gemini AI model and reinforcement learning to generate role-specific questions. It is used to apply psychological scaffolding (progressively increasing question complexity), fulfilling the role of structured interview preparation and confidence building. Addressing the highly technical nature of IT hiring, PS05 details a conversational AI (GPT-4o-mini) designed to support the "think-aloud" protocol required in coding interviews. This tool prompts candidates to verbalize their problem-solving processes while writing code, fulfilling the role of cognitive assessment and technical communication training. Complementing this, PS14 introduces VoxMentor, a voice-driven platform utilizing Whisper ASR, Google Gemini, and Judge0. VoxMentor is used to execute automated coding evaluations alongside company-specific question repositories (e.g., Google, Amazon), fulfilling the role of immersive technical simulation and highly targeted corporate recruitment preparation.

Beyond verbal interactions, multimodal neural networks are extensively applied to conduct objective behavioral and emotional assessments during IT interviews. PS11 highlights a real-time evaluation system powered by Convolutional Neural Networks (CNNs) that captures live video and audio streams. Used continuously during the mock interview, it analyzes facial expressions, speech tone, and body posture, fulfilling the role of objective behavioral assessment by scoring candidates on soft skills like attentiveness and stress management. Along similar lines, PS09 presents MockMate, which pairs CNNs (ResNet50) for facial emotion recognition with Recurrent Neural Networks (RNNs/LSTMs) for voice confidence analysis. It is used to process visual and prosodic features simultaneously, fulfilling the role of real-time emotional scoring and comprehensive performance evaluation. Expanding the scope of captured biometrics, PS10 applies LSTM and BiLSTM sequential regression models to acoustic, linguistic, gaze, and biological data (heart rate and electrodermal activity via wristbands). This tool is used during Virtual Reality Head-Mounted Display (VR-HMD) interviews, fulfilling the role of predicting true communication skills and estimating the psychological discrepancy (gap) between a candidate's actual performance and their self-efficacy. Furthermore, PS04 utilizes off-the-shelf recognition technology (such as AFFDEX and Nemesysco) to capture multimodal social signals. It is used to present trainees with an interactive emotion dashboard, fulfilling the role of providing automated non-verbal feedback to enhance essential media and communication skills.

Finally, immersive technologies, microlearning, and diagnostic frameworks are applied to bridge specific skill gaps and optimize the overall workforce readiness of IT candidates. PS13 utilizes immersive Virtual Reality (VR) to enable "body swapping," allowing users to answer questions from the perspective of an interviewee and subsequently review their performance from the embodied perspective of the recruiter. This tool is used to induce self-distancing, fulfilling the role of advanced soft-skill self-assessment and emotional regulation in high-pressure hiring contexts. To ensure candidates possess the baseline competencies required before the interview, PS07 examines the application of mobile-based microlearning interventions and gamification. Used as a pre-selection preparatory tool, it delivers bite-sized training tailored to technical and engineering disciplines, fulfilling the role of targeted soft-skill development (such as leadership and time management) to align student capabilities with corporate project management expectations. Assessing the overarching impact of these technologies, PS15 employs the Technology Acceptance Model (TAM) and Social Cognitive Career Theory (SCCT) to evaluate student engagement with LLMs like ChatGPT and Microsoft Copilot. This diagnostic approach is used to survey familiarity and usage patterns, fulfilling the role of identifying technological underutilization in career preparation and guiding the strategic integration of AI tools to effectively close the industry-academia gap.

C. *Common Communication Challenges Faced by Entry-Level Professionals*

The reviewed studies converge on a consistent and structurally coherent picture: entry-level IT professionals face a multidimensional set of communication barriers in recruitment processes that extend well beyond simple nervousness or lack of preparation. The evidence extracted from the fifteen selected studies allows these barriers to be organized into five thematic clusters: anxiety and evaluative threat, self-assessment deficit, technical-to-verbal translation failures, structural and institutional gaps, and psycholinguistic and behavioral deficiencies..

Anxiety and evaluative threat emerge as the most frequently documented barrier across the corpus. Studies PS01, PS02, PS03, PS05, PS11, PS13, and PS15 all identify anxiety as a core impediment to effective communication during interviews. PS03 provides the most methodologically rigorous evidence: using the State-Trait Anxiety Inventory (STAI), the study recorded mean pre-intervention anxiety scores that dropped by $45\% \pm 12$ in an AI-assisted group versus only $20\% \pm 15$ in a control group, establishing anxiety not merely as a motivational claim but as a measurable and differentially treatable communication barrier. PS13 extends this finding by demonstrating that standard first-person-only practice, the default training modality for most inexperienced candidates, produces significantly inferior self-awareness of non-verbal cues ($z = -4.28$, $p < 0.001$), suggesting that anxiety is compounded by an inability to observe one's own behavioral performance. PS02 adds important nuance, noting that IT students, whose profiles are typically more comfortable with code and screens than with face-to-face evaluation, experience a specific form of evaluative anxiety when confronted visually by an interviewer, impairing real-time logical structuring of responses.

A second and particularly consequential barrier is the self-assessment deficit, the inability of inexperienced candidates to accurately gauge their own communicative performance. PS10 provides the most quantitatively precise evidence: the Pearson correlation between expert-assessed communication skill (CS) and self-reported self-efficacy (SE) was $r = 0.124$, a value so low as to indicate near-total absence of correlation. This means that how well a candidate feels they performed is almost entirely decoupled from how an expert actually evaluates them. The consequences are systemic: candidates may reject accurate feedback, fail to prioritize areas of genuine weakness, or enter subsequent interviews with miscalibrated expectations. The same study reveals that biological markers of anxiety, heart rate and electrodermal activity, were stronger predictors of self-efficacy than any linguistic or acoustic feature, implying that the anxiety response operates at an involuntary, autonomic level that candidates cannot consciously regulate or observe in themselves. PS13 corroborates this finding from a different angle, showing that third-person visual perspective training (virtual reality body-swapping) significantly improved self-assessment accuracy for non-verbal behavior, reinforcing that the barrier is partly perceptual: inexperienced candidates simply cannot see themselves as interviewers see them.

The third cluster concerns technical-to-verbal translation failures, a barrier particularly salient for IT candidates. PS05 documents the specific cognitive load imposed by technical interviews requiring simultaneous problem-solving and verbal narration. The paper identifies what it terms the "think-aloud" challenge: candidates struggle not only with what to say but with when to say it relative to their own coding process. Participant testimony explicitly captures this: "figuring out when I should be thinking and actually coding versus actually like explaining" (P17, PS05). This dual-task demand induces anxiety and produces the phenomenon of "dead air", prolonged silences that interviewers may interpret as lack of competence even when the candidate is cognitively engaged. PS06 frames the same barrier in broader terms: IT undergraduates frequently struggle to translate their technical knowledge into verbal communication, leading to grammatical gaps, fluency failures, inconsistent tone and formality, and non-verbal inconsistencies such as avoidance of eye contact and inappropriate facial expressions. PS09 adds a related and underappreciated failure mode: the reliance on memorized responses. Inexperienced candidates tend to rehearse scripted answers and break down when interviewers probe technically or demand dynamic, improvised reasoning, revealing that surface-level preparation substitutes for, rather than develops, genuine communicative competence.

The fourth cluster consists of structural and institutional barriers that predispose candidates to enter recruitment processes underprepared. PS07, PS12, and PS15 each address, from different angles, the academic-industry gap that leaves graduating IT professionals without adequate communication training. PS07 documents this empirically: pre-intervention communication skill scores across all university disciplines averaged $M = 53.21$ ($SD = 8.45$), representing only moderate baseline competency, with engineering and technology students, the group most analogous to IT candidates, showing the largest absolute gain from a structured soft-skills intervention ($\Delta M = +7.42$), suggesting this group enters the market with the most pronounced communication deficit. PS15 reinforces this structurally, reporting that 96% of employers perceive graduating candidates as insufficiently prepared, and that the majority of students do not actively use available tools to develop communication competencies (only 36% rated their agreement with AI-assisted communication development at 4 or 5 on a 5-point scale). PS15 also identifies an equity dimension: candidates from underrepresented groups face additional institutional barriers that limit access to high-quality career preparation experiences, amplifying pre-existing communication deficits.

The fifth cluster encompasses psycholinguistic and behavioral deficiencies that manifest in observable communicative output. PS04 documents that, even among participants with some public speaking background, pre-training interviews exhibited significant prevalence of negative facial expressions, frowning and disgust, and that candidates exposed to awareness of their own performance without structured feedback experienced significant confidence reduction. PS14 highlights vocal modulation and articulation as specific failure points: inexperienced IT candidates often lack the capacity to verbalize technical knowledge with an appropriately confident vocal tone, resulting in vague responses, constant hesitations, and a flat, monotonous delivery under evaluative pressure. PS11 operationalizes these deficits through grammar, vocabulary, and fluency as measurable dimensions of communication performance, while PS03 identifies four discrete dimensions, clarity, conciseness, coherence, and overall articulation, along which inexperienced candidates show measurable gaps even when they believe their performance to be adequate.

It is notable that three studies, PS08, PS12, and, to a significant extent, PS04 do not substantively address Q2. PS08 operates at the system-design level, modeling an idealized candidate rather than studying real human communication behavior; PS12 analyzes job advertisement texts rather than candidate experience; and PS04, while providing some tangential observations, does not target IT professionals or hiring contexts. These absences are structurally expected given each study's scope and publication venue, but they represent gaps in the corpus that future research would benefit from addressing through direct, empirical investigation of candidate-side communication failures within IT-specific hiring scenarios.

In synthesis, the evidence across the reviewed studies confirms that communication barriers in IT recruitment are not reducible to a single cause. They are simultaneously psychological (anxiety, low self-efficacy, miscalibrated self-assessment), cognitive (dual-task overload, reliance on memorized scripts), behavioral (vocal, non-verbal, and articulatory deficiencies), and structural (academic preparation gaps, inequitable access to mentorship). Effective interventions must therefore operate across all these dimensions, a conclusion that directly motivates the investigation of AI-powered tools and structured communication strategies addressed in the subsequent subsections of this review.

D. *Benefits and Risks Associated with AI-Mediated Recruitment*

The evidence synthesized under Q3 converges on a dual-edged portrait of AI-powered tools and structured communication strategies in IT recruitment: the same mechanisms that make these systems scalable, consistent, and instrumentally effective also generate the limitations that circumscribe their use. Empirical results across the corpus report gains in communication performance, interview confidence, self-assessment accuracy, and two-sided matching quality, yet the same studies document residual weaknesses in empathy, content-quality judgment, ecological validity, and data sensitivity. Rather than arbitrating between enthusiastic and cautionary framings, this section develops four interlocking themes that emerge from the corpus — emotional regulation and self-efficacy, metacognitive calibration, objectivity and scalability, and the limits of automated assessment — and closes with the conditions under which these tools appear to produce net-positive outcomes in IT hiring contexts.

A first cluster of evidence concerns the affective dimension of interview preparation. Job interviews are characteristically associated with high anxiety, frustration, and distress rooted in their evaluative and competitive nature (PS13), and interventions that scaffold emotional regulation appear to translate into measurable performance gains. Virtual-reality body-swapping protocols that induce self-distancing through third-person perspective (3PP) exploit the Proteus effect and reduce the negative ruminations that first-person perspective (1PP) tends to amplify (PS13). In parallel, automated multimodal feedback on nonverbal social signals raises post-training confidence and skill ratings relative to traditional feedback, while the traditional-feedback condition shows a significant reduction in confidence from pre- to post-training — a non-trivial negative impact of the incumbent approach (PS04). PS03 similarly reports significant improvements in communication skills, confidence, and overall interview performance for an AI-supported group. Conversely, the same body of work flags a ceiling: AI feedback, however structured, lacks the human empathy needed to address deep-seated anxieties and motivational concerns (PS03), suggesting that emotional gains are real but bounded.

A second, closely related theme is metacognitive calibration — the alignment between demonstrated skill and self-perceived competence. PS10 frames this explicitly through the estimation of a "gap" (GA) between communication skill and self-efficacy, arguing that automated systems can detect anxious or underconfident candidates and then tailor feedback so that, for instance, highly skilled but low-efficacy interviewees are nudged toward improved self-impression rather than additional skill drilling (PS10). PS13 reports analogous effects: Out-of-Body and Recruiter conditions outperform a No-Swap baseline precisely on self-assessment of the communication styles used during the interview, indicating that embodiment-based interventions change what trainees notice about themselves, not only what they do. PS04 contributes longitudinal corroboration through a six-month follow-up in which neutral observers rated the social-signal-feedback group as significantly better communicators, with a 20% versus 15% subjective improvement over controls — evidence that metacognitive recalibration persists beyond the training session. The recurring mechanism across these studies is that AI mediation externalizes otherwise tacit performance signals, making them available for reflection.

A third theme concerns the organizational and systemic impacts of AI in recruitment: objectivity, scalability, accessibility, and bias mitigation. PS11 argues that CNN-based facial-emotion detection enables objective evaluation without human bias and enhances the accuracy and reliability of interview assessments while saving time and effort. PS10 makes the complementary argument on the employer side, proposing that automated systems can handle low-level, first-round interviews and thereby reduce rating blur caused by interviewer subjectivity (PS10). At the matching-infrastructure layer, PS08 demonstrates that a multi-agent LLM framework improves person–job matching accuracy and adaptability across domains, with statistically significant two-sided matching gains and measurable — if partial — reductions in gender-linked disparities compared to baselines (PS08). Scalability and access compound these effects: PS03 describes how digital platforms democratize employability training in resource-constrained environments, and PS15 converges on the view that AI tools "can help scale up career preparation" by exposing students to scenarios closer to industry practice (PS03; PS15). Microlearning interventions extend this logic to curricular delivery, producing measurable enhancements in specific soft skills when embedded into university programs (PS07).

The same corpus, however, delineates the frontier where these gains attenuate. PS15 concedes that current

AI does not yet fully assess the substantive quality of answer content without preset criteria — a decisive limitation in technical IT interviews where correctness and reasoning depth dominate. PS11 documents that existing models suffer from weak multimodal integration, sensitivity to environmental conditions, and high computational cost, with hybrid approaches struggling to adapt to user diversity and emotional variation (PS11). PS03 further raises a data-governance concern that is acute for IT contexts: simulations capture sensitive personal details including career aspirations, identity information, and behavioral responses, creating privacy exposure that a purely pedagogical framing tends to underplay. Fairness gains, too, are qualified rather than absolute — PS08 reports that bias is reduced relative to baseline, not eliminated, and disparities across genders persist within the system's outputs. PS07 adds a pedagogical caveat whose force extends to interview-training platforms built on short, decomposed interactions: when content is broken into small segments, the risk of oversimplification rises and gaps in understanding can emerge (PS07).

Taken together, the corpus suggests that AI tools and structured communication strategies produce net-positive impacts in IT recruitment under a relatively specific set of conditions: when they are positioned as a supplement to, rather than a substitute for, human judgment and traditional training (PS15); when they target affective and metacognitive dimensions — anxiety, self-efficacy, nonverbal awareness, gap estimation — where externalized signals genuinely outperform introspection (PS13; PS04; PS10); when content-substance evaluation is either scaffolded by preset criteria or retained by human interviewers (PS15); and when the systemic gains in scalability, matching quality, and bias reduction are paired with explicit attention to data sensitivity and residual fairness gaps (PS08; PS03). Outside these conditions, the same tools risk displacing the judgment they were meant to augment, or encoding the subjectivities they were meant to correct. The dialectic is not between adoption and rejection but between configurations of AI mediation that empirically amplify candidate capability and those that do not.

E. *Effective Communication Strategies and Training Approaches*

This section synthesizes the most effective structured communication techniques and training methodologies identified in the literature. While this systematic mapping analyzed 15 primary studies, the following discussion highlights six specific works—PS14, PS02, PS09, PS08, PS04, and PS10. These studies were selected as the core evidence for this section due to their high methodological rigor in evaluating real-time interaction frameworks, adaptive interviewing engines, and multimodal feedback systems, providing actionable strategies for entry-level IT professionals.

The first major strategy identified focuses on the transition from rigid, scripted responses to dynamic logical articulation. Inexperienced candidates often rely on memorized scripts, which fail during technical deep-dives or unexpected follow-up questions. To mitigate this, the literature proposes the Adaptive STAR method (Situation, Task, Action, and Result), where the candidate is trained to provide information in modular blocks that allow for interruptions and a deeper level of detail (PS09). This is further complemented by the "Think Out Loud" technique, a strategic approach that requires candidates to vocalize their problem-solving process in real-time. According to PS08, this technique is vital for technical interviews as it transforms silence into a transparent display of cognitive competence, allowing recruiters to assess the candidate's reasoning path even if the final solution is not immediately reached.

Beyond verbal logic, the integration of non-verbal signals emerges as a critical training approach. For IT students who frequently encounter social anxiety during evaluative interactions, Systematic Desensitization through immersive role-playing with AI-powered MetaHuman avatars proved to be a superior training strategy (PS02). This approach enables candidates to practice maintaining eye contact and open posture in a "safe-to-fail" environment. As PS04 observes, the key objective is achieving Non-Verbal Alignment, ensuring that the candidate's physical cues—such as facial expressions and hand gestures—are congruent with their technical explanations, thereby projecting professional maturity and leadership potential.

Acoustic clarity and vocal control also constitute a fundamental pillar of effective communication. Through voice-driven AI simulations, candidates can apply Conscious Vocal Modulation as a strategic tool to sound more authoritative. This involves the deliberate suppression of linguistic fillers (e.g., "uhm," "like") and the control of speech cadence. PS14 emphasizes that being able to maintain a steady, confident tone during complex architectural discussions is interpreted by recruiters as a proxy for technical mastery. The effectiveness of this training is significantly enhanced when simulators allow for Company-Specific simulations, preparing the student for the unique communication styles and cultural expectations of specific tech giants.

Finally, the overarching framework for these strategies is the Evidence-Based Iterative Practice, supported by multimodal feedback. Rather than relying on subjective self-assessment, learners utilize objective data—such as eye-contact percentages and vocal pitch stability—to guide their development. By treating communication as a measurable output, akin to a software debugging cycle, students can perform targeted adjustments to their performance (PS10). This iterative loop, fueled by AI diagnostics, ensures that soft skill acquisition is not an abstract goal but a precise, data-driven enhancement of the professional's total profile.

V. Integrated findings

The comprehensive synthesis of the 15 primary studies selected for this systematic mapping provides a holistic view of the intersection between technological innovation and the development of soft skills for entry-level IT professionals. The evidence indicates that the recruitment landscape is undergoing a paradigm shift, moving away from static preparation toward adaptive, immersive, and data-driven ecosystems. A core finding across the literature is that Artificial Intelligence (AI) and Virtual Reality (VR) act not merely as assessment tools, but as critical catalysts for psychological and behavioral readiness. For instance, the integration of Embodied AI through MetaHuman avatars and VR body-swapping techniques offers a "safe-to-fail" environment that is essential for desensitizing candidates to the social stress of evaluative interactions (PS02; PS13).

This technological evolution is directly linked to the characterization of the "Perception-Performance Paradox" prevalent in the computing industry. While market data reinforces that communication is the most demanded collaborative skill (Diniz et al., 2024), there is a documented temporal delay in how junior professionals recognize and value these competencies (Ivory et al., 2024). This gap manifests in high levels of evaluation anxiety and a lack of non-verbal congruency during technical "whiteboard" interviews. The integrated data suggests that technical mastery alone is insufficient; successful candidates are those who can effectively articulate their logic under pressure, a skill that the mapped studies show can be systematically enhanced through adaptive interviewing engines that utilize non-linear questioning (PS08; PS09).

Furthermore, the research confirms the superior efficacy of multimodal and iterative feedback over traditional peer-to-peer practice. By utilizing automated recognition of non-verbal signals and acoustic analysis, candidates are provided with objective metrics—such as vocal pitch stability, eye-contact frequency, and speech cadence—that transform soft skill acquisition into a measurable "debugging cycle" (PS10; PS04; PS14). This approach empowers inexperienced professionals to move beyond subjective self-assessment, allowing for precise, evidence-based adjustments to their professional persona. However, the findings also raise a critical cautionary note regarding the risk of dehumanization and algorithmic bias in recruitment processes (Ratnesh et al., 2026). The consensus among the analyzed studies is that the most effective training model is a hybrid one: leveraging AI for intensive simulation and objective diagnostics while maintaining human oversight to ensure ethical integrity and cultural alignment.

VI. Final remarks

This Systematic Mapping Study investigated how the strategic integration of AI-driven tools and structured communication methodologies can enhance the recruitment outcomes for entry-level IT professionals. The synthesis of 15 primary studies reveals that effective communication is no longer a peripheral trait, but a non-negotiable strategic competency required to transform technical potential into organizational value. Despite this, junior professionals frequently experience a profound soft skills gap characterized by high evaluative anxiety, severe self-assessment deficits, and the cognitive overload of translating complex technical knowledge into verbal articulation.

To bridge this persistent industry-academia divide, Artificial Intelligence and immersive technologies are functioning as transformational engines. Systems leveraging Large Language Models (LLMs), multimodal neural networks (CNNs/RNNs), and Virtual Reality (VR) provide inexperienced candidates with a risk-free environment for rigorous practice. By supplying objective, automated feedback on verbal fluency and non-verbal metrics, such as vocal pitch stability and eye contact, these technologies allow candidates to treat soft skill acquisition as a measurable, iterative debugging process. When paired with structured methodologies like the Adaptive STAR method and the Think-Aloud protocol, candidates are empowered to transition away from rigid, memorized scripts and toward dynamic logical articulation.

Despite these measurable benefits, the reviewed literature explicitly cautions against the unmitigated adoption of automated assessment. Current AI tools inherently lack the human empathy needed to resolve deep-seated anxieties and struggle to fully evaluate the substantive quality of complex technical reasoning without preset criteria. Furthermore, systemic risks regarding data privacy, algorithmic bias, and the potential for oversimplification remain critical barriers. Consequently, the most effective paradigm for modern recruitment preparation is a hybrid model: one that leverages AI for scalable simulation, objective diagnostics, and metacognitive calibration, while fundamentally retaining human oversight to ensure cultural alignment, substantive evaluation, and ethical integrity.

Moving forward, this study highlights several essential avenues for future research. First, there is a significant geographical gap in the current literature, with an absence of studies originating from Latin America and Africa, necessitating future research to explore these diverse labor markets. Second, the field would benefit from direct, empirical investigations of candidate-side communication failures in real-world, IT-specific hiring scenarios, rather than relying strictly on simulated models or job advertisement text mining. Addressing these limitations will be vital to guarantee that technological interventions serve responsibly as a developmental ally rather than a crutch for the next generation of global IT talent.

VII. Acknowledging

It is acknowledged that AI-assisted tools, including large language models, were used to support language refinement in certain sections of this paper, improving clarity and readability, while all analytical and interpretive decisions were made exclusively by the research team.

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